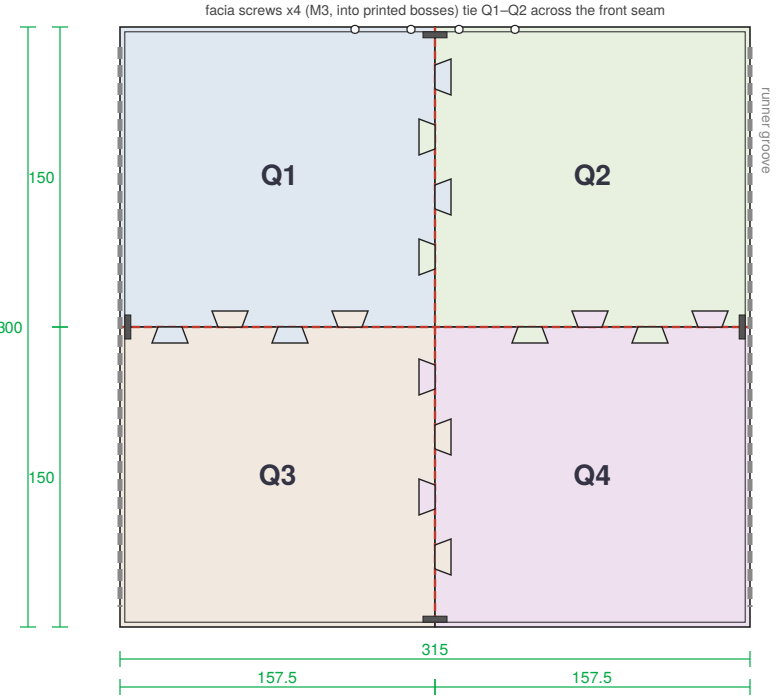
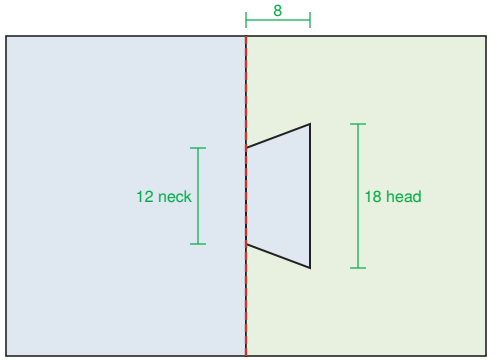
BOTTOM DRAWER — FOUR-QUARTER BUILD (D2 option a) v1

Plan, 1:1, facia removed. Drawer 315 x 300 x 60 stays full size; printed as four quarters, each 157.5 x 150 (flat on the 256 bed). Q1 front-left … Q4 rear-right.



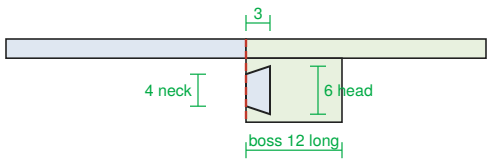
FRONT is the top edge of the plan. Red dashes = seams. Coloured tabs = floor puzzle-dovetails, owner shown by colour, alternate along each seam, first at 25 from the wall, pitch 30, none within 15 of the cross.
Grey blocks = female bosses of the wall dovetails (front, rear, left, right walls). Runner grooves on the outer side walls cross the rear seam; the wall dovetail keeps them aligned.

DETAIL A — floor puzzle-dovetail (plan), 4:1



Floor 3.0 thick. Tab clearance 0.15 all round (fit.py DOVETAIL_CLR) — coupon sets it. Tabs drop in from above, so every joint in the drer assembles vertically: Q2 onto Q1, Q3 onto Q1, Q4 onto Q2 + Q3 last. Layers lie in the floor plane, so tabs are full-strength in-plane.

DETAIL B — wall vertical sliding dovetail (plan section), 4:1



Wall 2.4. Rib on the Q1 wall end runs the full wall height (floor top to rim); pocket in a boss grown inboard on Q2 (boss 12 x 8 x full height).
Q2 slides down onto Q1: the dovetail takes the pull-apart load across the seam and keeps the two wall faces flush, so the runner groove stays straight.
Four crossings: front wall (hidden behind the facia), rear wall, left wall, right wall. CA-glue the four dovetails only after the runner test passes.

P7 COUPON (redefined for D2) — one PETG plate, gates every split part

1. Centre cross: four 60 x 60 floor corners with the real tab pattern (3 mm floor, 2.4 walls on the outer edges) — tests tab fit, flushness and the 15 mm keep-out at the cross.
2. Wall crossing: two 40-long wall ends, 60 tall, one with the rib, one with the boss — tests the vertical slide, pull-apart and face flushness.
3. Beam section (D6): two 40-long ends of the 9 x 40 sled beam with the same dovetail scaled to the section — tests the sled split.

Pass: all three assemble by hand, hold without glue, faces flush within 0.2. Iterate DOVETAIL_CLR in 0.05 steps; expect 0.15–0.25 on the X1C.
Print flat, tabs up, 0.2 layers, 4 walls, no supports. About one hour of PETG.

ASSEMBLY

Q1 on the bench → Q2 down onto Q1 (front + rear wall dovetails, floor tabs) → Q3 down onto Q1 (left wall dovetail, tabs) → Q4 down onto Q2 + Q3 → facia screwed across Q1–Q2.
Runner test (P3) on the assembled drawer before any glue. Housing top above it is four loose tiles (clipped, P4), not part of the drawer.